Gotta Go Fast:
Applying Constant Force to Reach the Universal Speed Limit

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BACKGROUND

Special relativity, in essence, is the patch for our physics engine . The developers realized that their foundation, known as Newtonian mechanics, could not accurately decipher observations seen in the field. Initially, the developers were confident that their Newtonian rules and functions could define everything. Even when the system appeared to fail, they could find ways to work around the complication, staying within the physics engine that had be established. However, as we know, there came a time where the Newtonian physics engine failed to fully bring the events into the old system. Albert Einstein, who you may have heard of once or twice, eventually wrote that a new patch must be made for the current understanding of physics, as we had not fully constrained these observations under the Newtonian system. In what we now call Special Relativity, Einstein theorized that at speeds approaching the speed of light, there are new effects that we must discern, most of which are acknowledged or understood through the original release. While the conditions on which this relativity takes place are hard to reach on Earth, needing very high speeds, it turns out that Newtonian mechanics is an approximation of our updated mechanics at slower speeds. At slower speeds, the effect is negligible, which is why classical mechanics is so useful as a building point towards higher level physics.

Relativity, as you can tell from the name, is the concept that everything we see is relative to our reference frame. For example, if you picked up your water bottle and let go of it, you would see the bottle fall towards the ground. However, from the bottle's frame of reference, you and everything in your reference frame moved upwards. All objects, no matter what reference frame they are in, must move through time. If two objects are in different reference frames, one must be moving. So, in the bottle's reference frame, you are moving and the bottle is at rest. Since it moves through time but not space, every other reference frame, with respect to the bottle, must be moving. Fortunately, Newton's laws is still valid and are even unchanged in these reference frames, as long as they are inertial. By definition, an inertial frame is any reference frame that holds all laws of physics unchanged. As long as any reference frame is traveling at a constant speed with respect to an inertial reference frame, that moving frame is also inertial. This is also known as the First Postulate of Relativity. Additionally, the Second Postulate of Relativity states that in all inertial frames, the speed of light c in a vacuum is constant. Special relativity, which is what we are dealing with in this paper, focuses almost completely on inertial reference frames, while accelerating reference frames is included in General relativity. We only need to account for general relativity when Newtonian gravity is not accurate in what we predict.

When dealing with different inertial reference frames, we must take into account that measurements will for each frame. While I stated that objects must move through time, that does not constrain objects to move through time at the same rate. As objects move faster and faster, time moves slower and slower in them due to what is known as time dilation. The relation between the at-rest inertial frame and the moving inertial frame is

$$\Delta t = \frac{\Delta t^{'}}{\sqrt{1 - \frac{v}{c}^2}}, \tag{1}$$

where $\Delta t^{'}$ is the time in the observed reference frame and v is the velocity of the moving frame with respect to the observers frame. We see in equation 1 that at low speeds, the times for each reference frame are approximately equal. The additional factor in time dilation is a common occurrence, which is why it is known as

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}, \tag{2}$$

with β being an abbreviation for $\frac{v}{c}$. This relation is known as the Galilean transformation, which we will later see that is not correct. However, we will burn that bridge when we get there. In addition to there being time dilation, we have the effect known as length contraction. If we look at the simplest equation for position,

$$l = vt, \tag{3}$$

with l being the distance, an easy comparison between our time t and the approximated time of the moving reference frame $t^{'}$, we see that

$$l = \frac{l^{'}}{\gamma} \tag{4}$$

, where we see that $l^{'}$ will either less than or equal to l. This effect is known as length contraction. While this a good approximation, we want to be able to look at length contraction of the distance for a moving reference frame, not just the length of an object. We have what are known as the Lorentz transformations, where, for motion entirely in the x direction, we have that

$$x^{'} = \gamma(x - vt) \tag{5}$$

and

$$t^{'} = \gamma(t - vx/c^2), \tag{6}$$

where the y and z coordinate are unchanged as both axes are perpendicular to the relative velocity. We get equation 5 from multiplying the updating position equations by γ and equation 6 from taking the inverse Lorentz transformation for equation 5 and substituting one in transformation into the other.

Four vectors are used to describe reference frames in relativity because in addition to space, they include a component for time. Both position and momentum are often represented by four vectors, but for now we only care about the four-momentum, which we define as

$$p = (\gamma m\dot{x}, \gamma m\dot{y}, \gamma m\dot{z}, mc). \tag{7}$$

You might recognize the fourth component of the momentum. One of Einstein's most famous equations, is

$$E = \gamma mc^2, \tag{8}$$

where m is the mass of the object and E is the energy of an object. At rest, we have that $E = mc^2$, which is known as the rest energy since at rest $\gamma = 1$ We see that $p_4 = E/c$, which is why the four-momentum is also known as the momentum-energy four-vector. Conservation of energy still applies, but because of the nature of four-vectors, we get that

$$p_o \cdot p_o = p_f \cdot p_f, \tag{9}$$

with

$$p_o = (0, 0, 0, mc) \tag{10}$$

and

$$p_f = (\gamma m\dot{x}, 0, 0, E/c), \tag{11}$$

where we have the y and z axes being perpendicular to the direction of motion.

The last part of the background will be where I introduce the one and only; Force. While it is with us always, in relativity it is defined as

$$\boldsymbol{F} = \frac{d\boldsymbol{p}}{dt}. \tag{12}$$

This is not the same as the non-relativistic force because $\boldsymbol{p}$ is not the non-relativistic momentum. Additionally, when working with forces and momentum, we can rewrite γ to be dependent on the force applied on the object. Seeing as we have equation 9, we can plug in equations 10 and 11 to get

$$-(mc)^2 = (\gamma m\dot{x})^2 - (E/c)^2, \tag{13}$$

where $p_o \cdot p_o$ is negative due to how the time-component is parameterized in relativistic coordinates. Recalling that $E = mc^2$, we see that

$$\gamma = \sqrt{1 + \frac{Ft}{mc}}, \tag{14}$$

where we substituted Ft for the momentum $m\dot{x}$.

METHODS

Our goal is to investigate where a situation where a constant force F in the x direction is applied to an object of mass m. We will be taking a look at position and velocity of this object as it speeds up towards c, the speed of light. As we have for most situations we have seen dealt with this semester, we will investigate using the Lagrangian. As long as the potentials have no dependence on the velocity, we can use the original equation In this situation, we can set the potentials to 0 in order to determine the our relativistic T. Remember that

$$p_x = \frac{d\mathcal{L}}{d\dot{x}}. \tag{15}$$

We know p_x from equation 7, so we can multiple both sides of equation 15 by $d\dot{x}$ and integrate to get that

$$T = \frac{1}{2}\gamma m\dot{x}^2. \tag{16}$$

Now that we have T, we can write out the Lagrangian as

$$\mathcal{L} = \frac{1}{2}\gamma m\dot{x}^2 + Fx, \tag{17}$$

where F is our constant force and we have no potentials in this scenario. With this, we can find that our solution for the Lagrangian is

$$F = \frac{d}{dt}(\gamma m\dot{x}). \tag{18}$$

As we are looking for the velocity and the position, as opposed to the acceleration of the system, we can multiple both sides by dt, integrate, and solve for the velocity to get that

$$\dot{x} = \frac{Ft}{\gamma m}, \tag{19}$$

and if we integrate once more for the position, we get that

$$x = \frac{F}{m}\left(\frac{mc}{F}\right)^2 \left(\sqrt{(}1 + \left(\frac{Ft}{mc}\right)^2) - 1\right) \tag{20}$$

As for our non-relativistic solutions, with all initial conditions for positions, velocities, and accelerations being 0, we have that

$$x_{nr} = \frac{1}{2}\frac{Ft^2}{m} \tag{21}$$

and

$$v_{nr} = \frac{Ft}{m}, \tag{22}$$

neither of which take into account the different reference frames of the object and the observer.

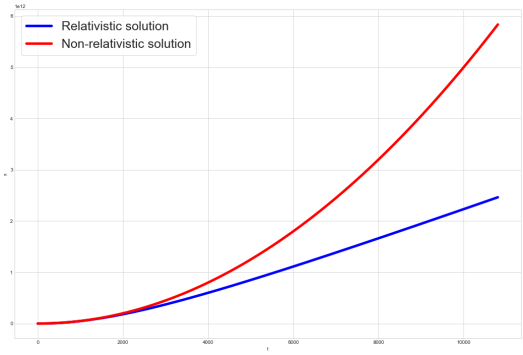

FIG. 1: The comparison between the relativistic and non-relativistic solution of $x(t)$ details that after some time the relativistic solution predicts a steady, linear increase of distance, while the non-relativistic travels at an exponential rate.

RESULTS

We plotted our resulting equations for the object of a time period of 3 hours. In figure 1, we see that the object would actually travel less than half as far as it would if it was only constrained by Newtonian mechanics. The γ factor limits the movement of the particle greatly, as the slope of the object's position eventually becomes constant. As we know that the slope of the position curve is the velocity, we can see that as the object approaches the speed of light, it begins to move in a linear fashion. As the non-relativistic solution does not account for different reference frames, it will continue to get faster and faster, moving at an exponential rate.

Figure 2, however, demonstrates the difference between our relativistic solution and the purely Newtonian solution. After .833 hours, the non-relativistic solution for our object exceeds the speed of light, whereas the relativistic solution slowly gets closer to the speed of light, but the object never truly reaches the speed of light. Which, as one might know, is impossible. Therefore, we can see that our relativistic solution correctly follows the laws of physics

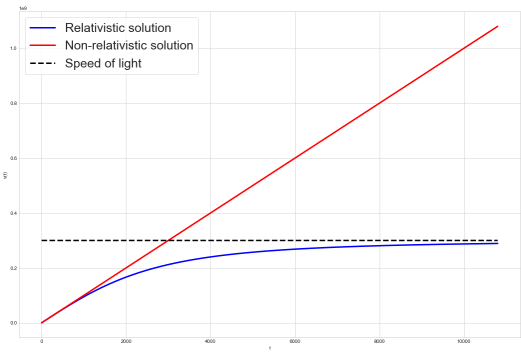

FIG. 2: Our relativistic solution demonstrates that it is impossible for anything to move faster than light. The